

Cryptocurrency Lead-Lag Statistical Arbitrage

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Research Motivation

Cryptocurrency markets exhibit strong cross-asset relationships due to shared liquidity pools, overlapping participant bases, and correlated macro drivers. However, price movements across assets are not always simultaneous.

Empirical observation suggests that **large movements in one asset propagate through the market with a short delay**, rather than occurring instantaneously. This raises the possibility that **lead-lag relationships exist between major cryptocurrencies** -- and if such relationships are measurable and persistent, they may provide a basis for statistical arbitrage strategies.

Research Hypothesis

Conditional on a large return shock in asset A at time t , the expected return of asset B at time $t + k$ is non-zero for small k .

Testing this requires:

- Synchronized historical price data across assets
- Event-based shock detection
- Cross-asset lag measurement and characterization
- Validation via realistic backtesting

Research Pipeline

The analysis is implemented as a staged notebook pipeline. Each stage produces artifacts consumed by the next, ensuring a **modular, reproducible, and auditable workflow**.

Stage	Notebook	Description
000	000_overview.ipynb	Structure, hypothesis, and design (this document)
001	notebooks/001_download.ipynb	Data acquisition -- Binance OHLCV for 9 crypto pairs
002	notebooks/002_enrich.ipynb	Feature engineering -- returns, rolling volatility, z-scores
003	notebooks/003_analysis.ipynb	Shock detection and cross-asset lag analysis
003a	notebooks/003a_regime_classification.ipynb	Regime segmentation, ADSR characterization, tradability filter
004	notebooks/004_strategy.ipynb	Signal construction and trade mapping
005	notebooks/005_backtest.ipynb	Backtesting, stress testing, and statistical validation
006	006_writeup.ipynb	Findings, conclusions, and future directions

Stage Definitions

Stage 1 -- Data Acquisition (001_download)

Reads from: Binance public API
Writes to: output/001_download/

Artifacts: PRICES.pkl , <COIN>.event_panel.pkl

- Defines the cryptocurrency universe (BTC, ETH, SOL, BNB, ADA, AVAX, XRP, DOGE, LTC)
- Downloads daily OHLCV price data
- Aligns time series across all assets
- Computes daily returns and constructs event panels

Stage 2 -- Data Enrichment (002_enrich)

Reads from: output/001_download/
Writes to: output/002_enrich/

Artifacts: returns_full.pkl , rolling_sigma.pkl , z_scores.pkl , price_wide.pkl

- Computes rolling volatility (sigma) for each asset
- Normalizes return signals as z-scores
- Prepares enriched datasets for event analysis

Stage 3 -- Analysis (003_analysis)

Reads from: output/002_enrich/
Writes to: output/003_analysis/

Artifacts: shock_events.pkl , extreme_z_scores.pkl , event_bitmap.pkl , sigma_event_matrix.pkl

- Identifies extreme shock events where |z-score| exceeds threshold

- Visualizes lagged cross-asset responses
- Characterizes the leader-follower relationship structure

Stage 4 -- Strategy Construction (004_strategy)

Reads from: output/002_enrich/ , output/003_analysis/
Writes to: output/004_strategy/

Artifacts: event_response_df.pkl , event_response_expanded.pkl , trade_df.pkl , daily_returns.pkl

- Converts observed lead-lag relationships into directional trade signals
- Defines trade construction logic (entry/exit, signal direction)
- Structures candidate trades for backtesting

Stage 5 -- Backtest (005_backtest)

Reads from: output/004_strategy/
Writes to: output/005_backtest/

Artifacts: trade_df.pkl , daily_returns.pkl , equity_curve.pkl , grid_results.pkl , performance_summary.pkl

- Simulates trade execution across a stress grid of execution delays and transaction costs
- Computes performance metrics: Sharpe ratio, win rate, drawdown
- **Statistical significance:** t-test on daily returns -> $t = 2.56$, $p = 0.011$
- **Sharpe confidence interval (Lo 2002):** 95% CI = [2.67, 3.86]
- **Walk-forward robustness:** Sharpe across 3 non-overlapping folds $\approx 4.43, 3.07, 2.66$

Stage 3a -- Regime Classification (003a_regime_classification)

Reads from: output/002_enrich/ , output/003_analysis/
Writes to: output/003a_regimes/

Artifacts: regime_series.pkl , adsr_profiles.pkl , regime_correlations.pkl , tradable_signals.pkl , context_signals.pkl , dragons.pkl

- Classifies every trading day into Bull / Bear / Transition / Dragon
- Characterizes ADSR profiles per shock type per regime
- Maps cross-asset correlations within each regime
- Applies the tradability filter to separate signal from context
- Documents Dragon events explicitly

Stage 6 -- Writeup (006_writeup)

Reads from: output/005_backtest/
Produces: Final research narrative

- Interprets backtest results against the original hypothesis
- Evaluates statistical robustness
- Documents limitations and proposes future directions

Scope

In scope:

- Major cryptocurrency assets on daily time resolution
- Short-term lead-lag relationships ($t+0$, $t+1$ execution)
- Statistical validation under realistic transaction cost assumptions

Out of scope:

- Intraday execution modeling
- Production deployment or live trading
- Portfolio-level optimization across multiple simultaneous strategies

Design Principles

1. **Determinism** -- each stage produces well-defined artifacts with no hidden dependencies
2. **Modularity** -- each stage can be executed and validated independently
3. **Reproducibility** -- all outputs are derived from persisted artifacts
4. **Transparency** -- all statistical assumptions are stated and tested explicitly

Summary

This document defines the structure, hypothesis, and execution plan for investigating lead-lag relationships in cryptocurrency markets. The pipeline implements, tests, and evaluates this hypothesis in a controlled and reproducible manner -- culminating in a validated statistical arbitrage strategy with strong and significant risk-adjusted returns.